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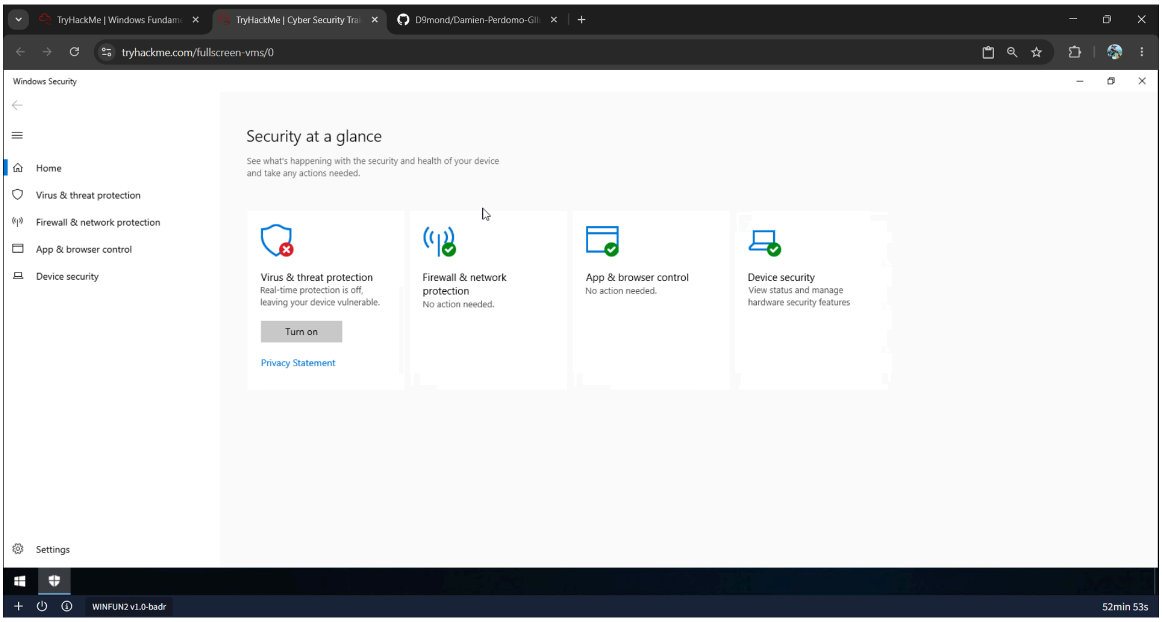
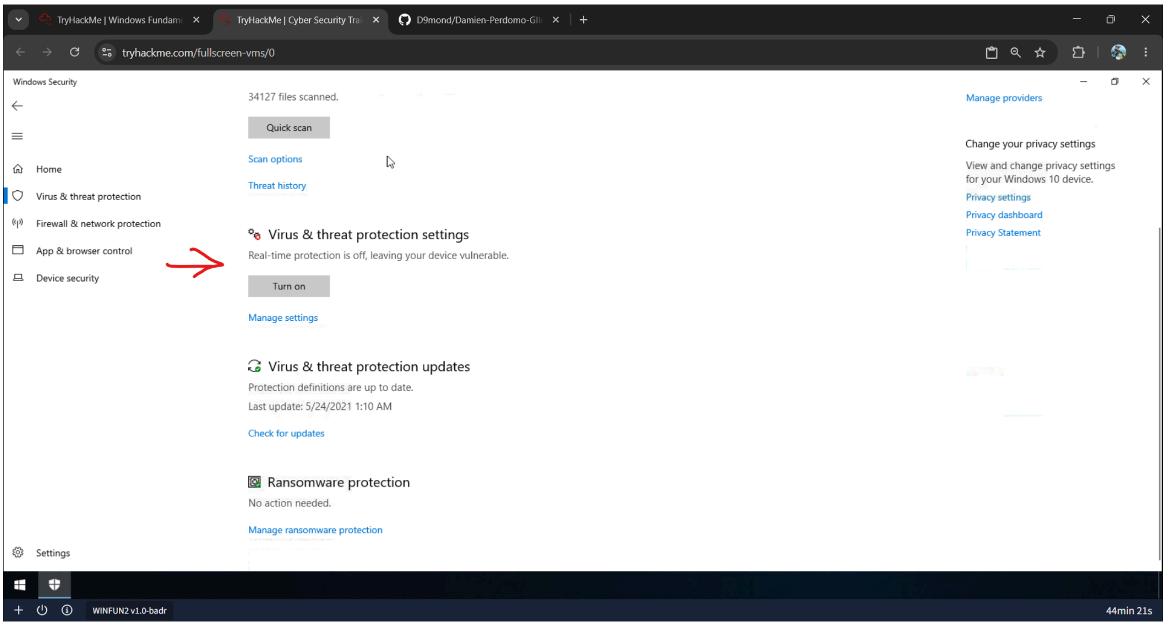
Environment: TryHackMe Windows Fundamentals 3 Lab
Focus Area: Windows Security Monitoring, Authentication Review & Operational Analysis

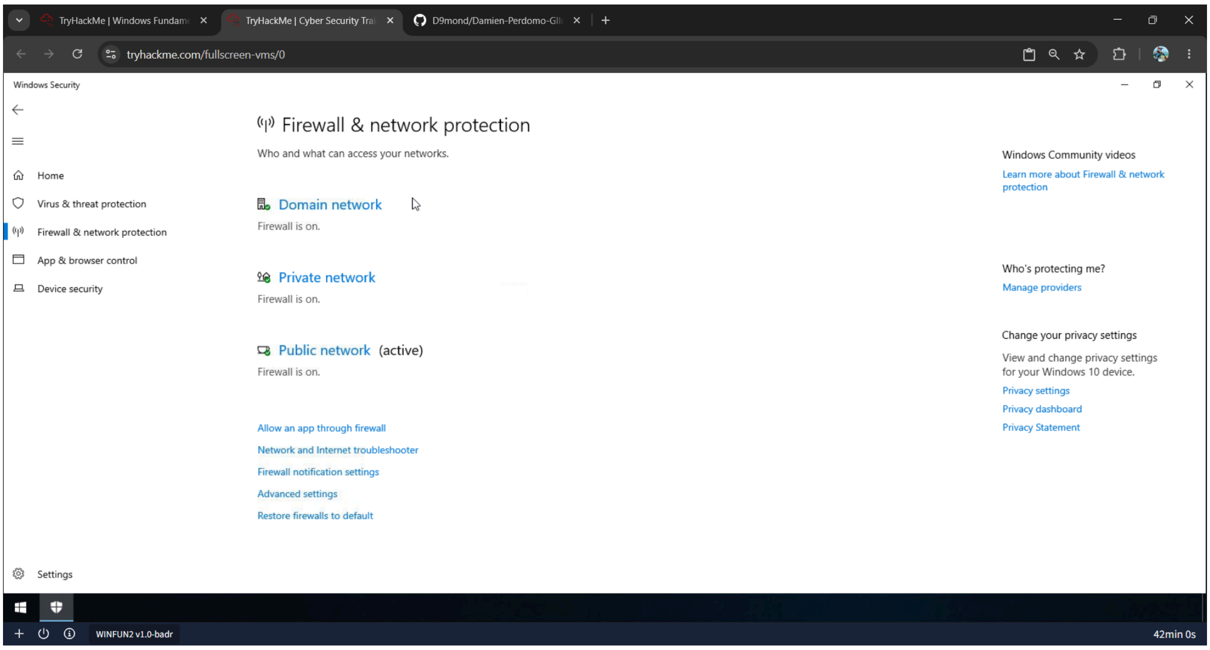
Windows Security Monitoring & Authentication Investigation



Overview
The Windows Security dashboard was reviewed to identify endpoint protection status, security recommendations, and operational security monitoring indicators within the Windows environment.

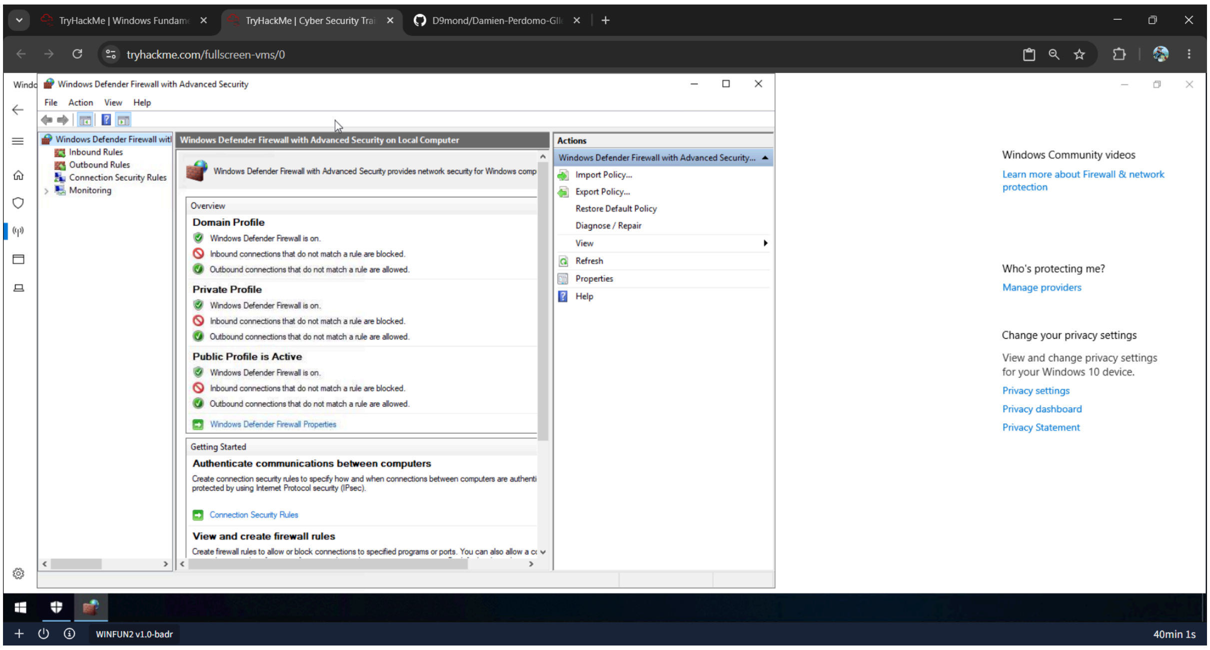


Real-time protection was identified as disabled within the lab environment. In production systems, disabled endpoint protection could increase exposure to malware execution and malicious activity.



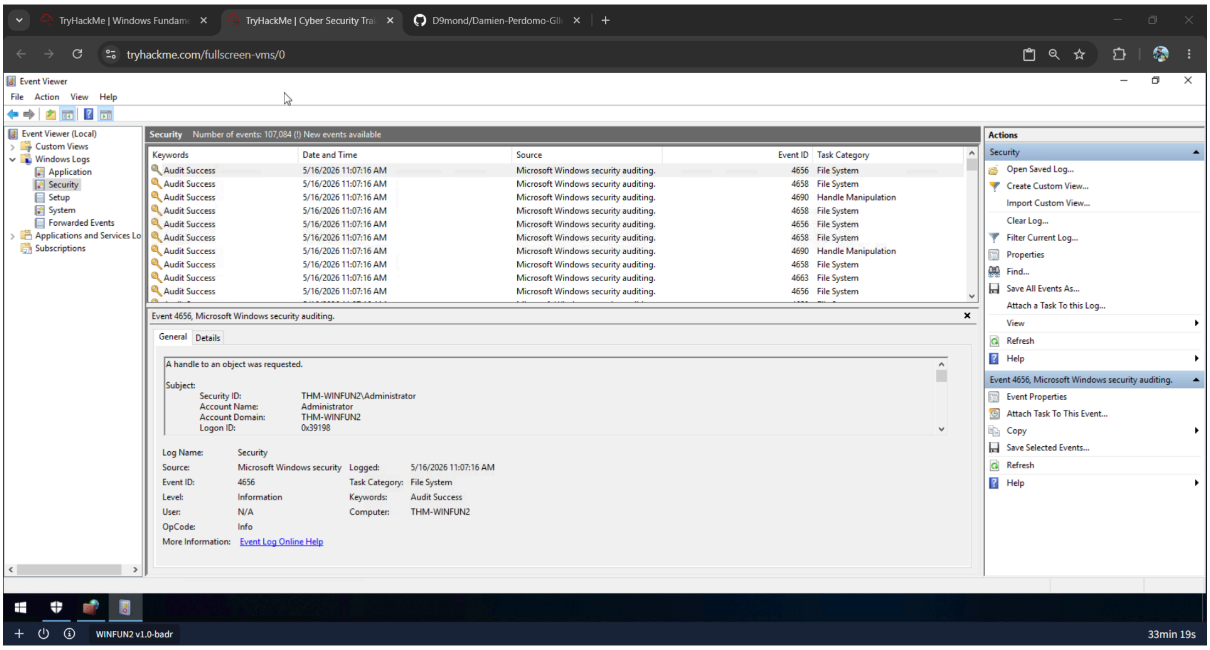
Investigation Process

Firewall profiles were reviewed to understand how Windows applies different security configurations depending on network trust level and exposure context.

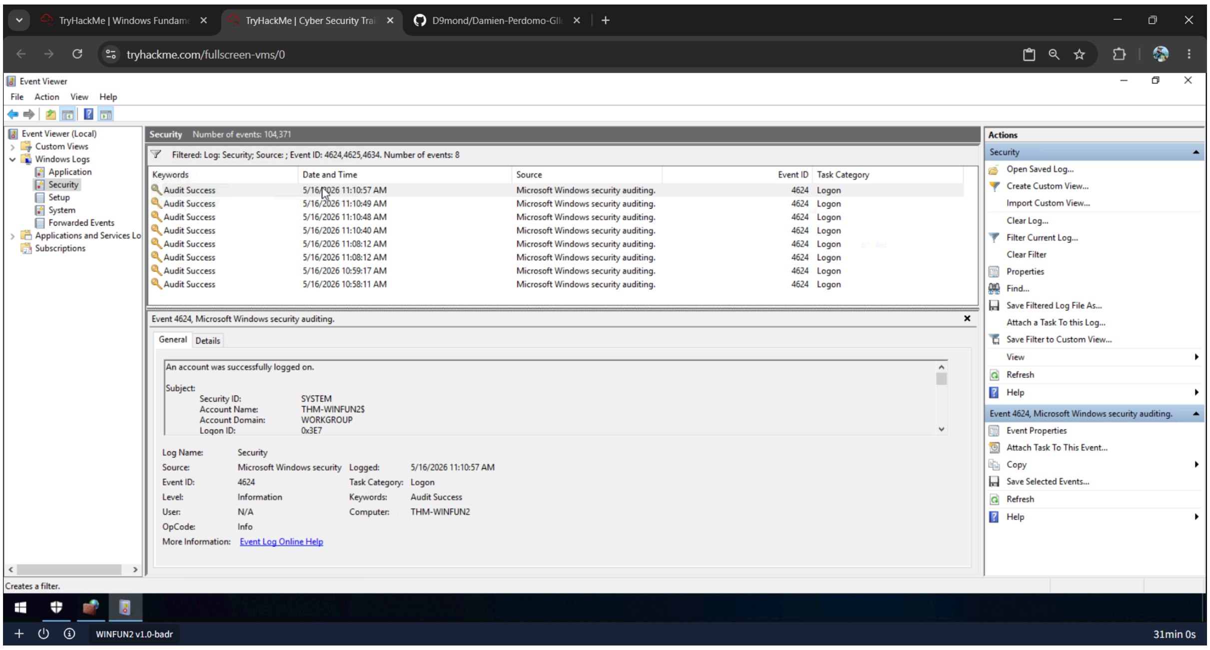


The Windows Defender Firewall advanced console was reviewed to analyze inbound and outbound filtering configurations, operational network controls, and host-level traffic management capabilities.

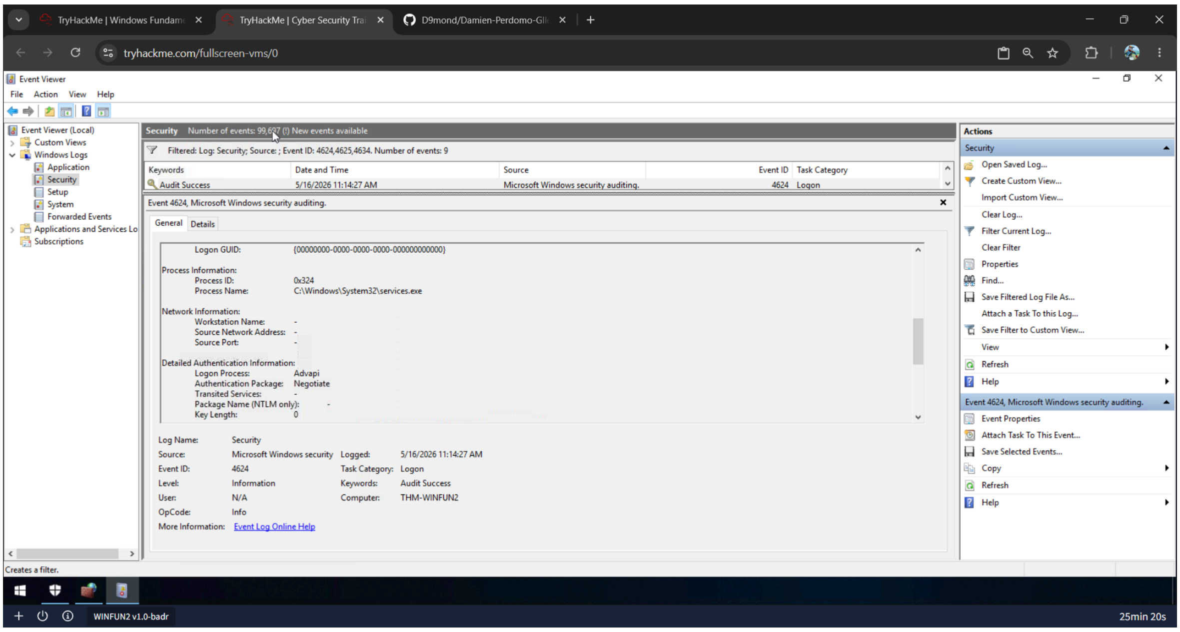
Authentication Log Review



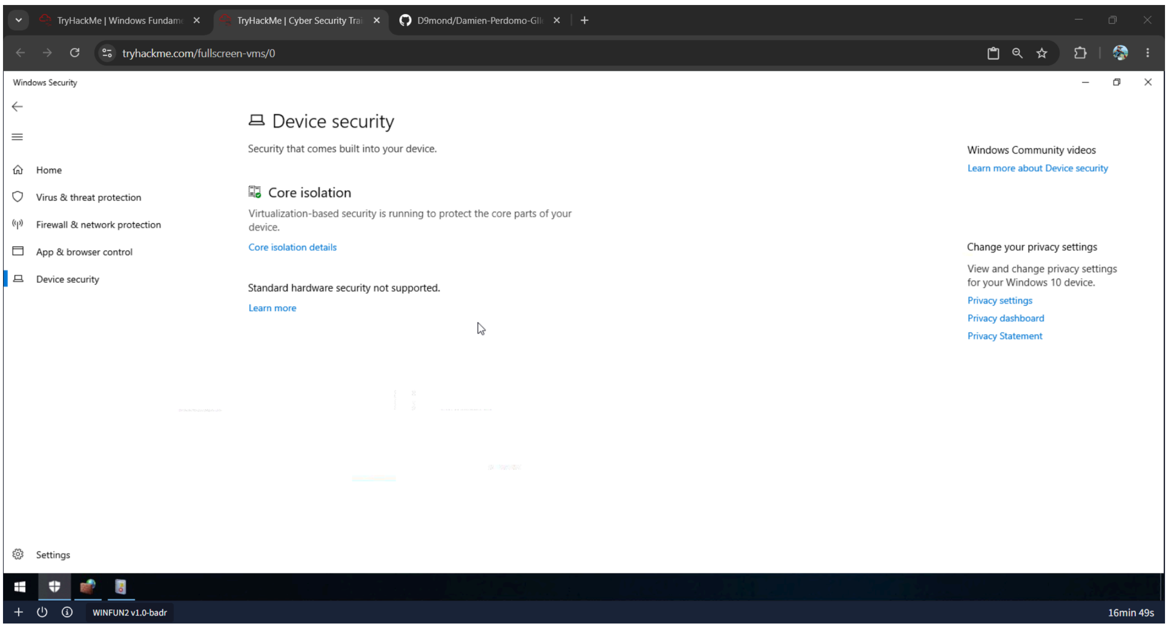
Windows Security logs were reviewed using Event Viewer to analyze authentication activity, security events, and operational monitoring data generated by the system.



Authentication-related Event IDs were filtered to review successful logons, failed authentication attempts, and session termination activity.



Authentication event details were reviewed to analyze account activity, logon behavior, timestamps, and workstation information relevant to operational monitoring and troubleshooting workflows.



Device security settings were reviewed to understand hardware-based protections and built-in Windows security mechanisms designed to improve endpoint protection and platform integrity.

Findings

- Real-time protection was disabled within the lab environment.
- Windows Firewall profiles were active and segmented by network type.
- Event Viewer provided visibility into authentication-related activity.
- Security logs contained operationally relevant authentication events.
- Windows built-in security tooling supports monitoring and troubleshooting workflows.

Operational Recommendations

- Maintain endpoint protection enabled in production environments.
- Monitor authentication-related Event IDs for repeated failures or abnormal behavior.
- Review firewall configurations based on network trust level.
- Use Event Viewer and security logs during troubleshooting and monitoring operations.

Key Takeaways

This investigation demonstrated how Windows built-in security tools can support technical troubleshooting, endpoint monitoring, authentication analysis, and operational security review activities within Windows environments.